EE1007
Introduction to Computer Science Laboratory

Lab 14
Class

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Lab 14.1
Elementary Arithmetic

- Elementary arithmetic includes the operations of addition(+), subtraction(-), multiplication(*), division(/) and modulo(%).
- You are asked to implement a program to calculate a given statement.
- This program will ask the user to enter a sequence of numbers and operators, and the sequence doesn't include any spaces and parentheses/brackets.
- Finally, the program will show the result after calculation.

Lab 14.1
Elementary Arithmetic

- For the order of arithmetic operations, multiplication, division and modulo are granted a higher precedence than addition and subtraction.
- All numbers in the sequence are non-negative integers with range[0, 2^31-1].
- In addition, the result of the calculation will be an integer in the range of [-2^31, 2^31-1].
- For division computation, the quotient must be an integer and the fraction should be unconditionally discard.
- In the program, the type of the variable storing the sequence must be string or char array. Otherwise you will get 0 point.

Lab 14.1
Elementary Arithmetic

- The screen dialog should appear as follows:
- Example 1

```
Please enter the arithmetic: 1+2+3+4+5+6/2
18
Process returned 0 (0x0)   execution time : 15.911 s
Press any key to continue.
```

- Example 2

```
Please enter the arithmetic: 1*2*3*4*5*6%2
0
Process returned 0 (0x0)   execution time : 16.676 s
Press any key to continue.
```

- Example 3

```
Please enter the arithmetic: 1+2+3+4+5+6%5
16
Process returned 0 (0x0)   execution time : 12.475 s
Press any key to continue.
```

Lab 14.2
Calculating the Number of Digits

- In this lab, you are asked to write a program for calculating the number of digits in a string, and replace all digits with "*" symbol.
- The program will ask the user to enter a string, and the string includes spaces, numbers, alphabets and special symbols.
- Considering the type of the variable in this program, you must use structure and string(str1) to store the input from the user.

```
struct example
  {
     string str1;
     string str2;
  };
```


Lab 14.2
Calculating the Number of Digits

- The program will calculate the number of digits in the string and showing the result on the screen.
- Finally, taking another string(str2) to store the string(str1) in the structure, and converting the digits to * in the string(str2). Then, show the second string(str2) in the program.

Lab 14.2
Calculating the Number of Digits

- The screen dialog should appear as follows:
- Example 1

```
Please enter a string.
abc123def456ghi789jk0
str1 = abc123def456ghi789jk0
The number of the digits in the str1 is 10.
str2 = abc***def***ghi***jk*

Process returned 0 (0x0)   execution time : 19.133 s
Press any key to continue.
```


Lab 14.2
Calculating the Number of Digits

- The screen dialog should appear as follows:
- Example 2

```
Please enter a string.
123 456 789 zxcvbnm,./
str1 = 123 456 789 zxcvbnm,./
The number of the digits in the str1 is 9.
str2 = *** *** *** zxcvbnm,./

Process returned 0 (0x0)   execution time : 18.368 s
Press any key to continue.
```

Lab 14.3

NCU Theater Reservation System(VI)

- Since more and more students have become a member of the NCU theater, it is getting much more difficult to manage the membership for the theater manager.
- Based on the previous system you had implemented in Lab 13.3, we would like to add some new features in our NCU theater reservation system.
- In this problem, you are asked to develop a function can **find member data** from his/her name in the system.
- You have to list all the information corresponding to the people with the name **exactly or partially** matches the searching keyword.

Lab 14.3

NCU Theater Reservation System(VI)

- Example
  - 3_1.txt

Name	Password	Phone
Andy	12345678	0975817468
Kevin	227788	0912345678
Jinx	456456	0812345678
Akali	77777	0812345678
Jax	878787	9032165498
Andy666	123456	1234567890
hiAndyisme	666666	6666666666
iamAndy	654321	0987654321

Lab 14.3
NCU Theater Reservation System(VI)

◆When you **use the keyword "Andy" for searching**, the outcome on the screen should be as follows

```
*** Customer Information ***
Name        Password    Phone
Andy        12345678    0975817468
Andy666     123456      1234567890
hiAndyisme  666666      6666666666
iamAndy     654321      0987654321
```


Lab 14.3
NCU Theater Reservation System(VI)

◆ The main menu of your system is shown as follows:

Before

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to exit the system
Enter your selection Here:
```

After

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to find member data
Please type 10 to exit the system
Enter your selection Here:
```

Lab 14.3
NCU Theater Reservation System(VI)

◆ The main menu of your system is shown as follows:

Before login

After login

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to find member data
Please type 10 to exit the system
Enter your selection Here: 5

***NCU Theater Member Center***
Do you have the membership? (y/n) y
Enter Username: yes123
Enter Password: 123

*** Welcome to NCU theater reservation system ***
***Welcome back yes123***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to log-out
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to find member data
Please type 10 to exit the system
Enter your selection Here:
```


Lab 14.3
NCU Theater Reservation System(VI)

◆The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to find member data
Please type 10 to exit the system
Enter your selection Here: 7

***Load member information***
Please enter the member file name: 3_1.txt
The file 3_1.txt has been loaded...
```

Lab 14.3
NCU Theater Reservation System(VI)

- The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to find member data
Please type 10 to exit the system
Enter your selection Here: 9

***Find member data***
Please enter the search keyword: Andy

The searching keyword is "Andy"...
The searching results are as follows:
*** Customer Information ***
Name       Password       Phone
Andy       12345678       0975817468
Andy666    123456         1234567890
hiAndyisme 666666         6666666666
iamAndy    654321         0987654321

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
```


Lab 14.3
NCU Theater Reservation System(VI)

- First, the program will print the message below.
  - "***Find member data***"
- Then, the program will ask what is the keyword that user wants to find.
- After user enters the keyword, the program should print the message below.
  - "The searching keyword is "(keyword)"…"
  - "The searching results are as follows"
  - "*** Costumer Information***"
  - …

Lab 14.3
NCU Theater Reservation System(VI)

- Requirement:
  - After finding the data of members, the program will return to the main menu.
  - If the user enters a keyword that doesn't match any data of the members, show the message **"We cannot find the keyword"** and return to the main menu.

Lab 14.3
NCU Theater Reservation System(VI)

- The screen dialog should appear as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to load member information
Please type 8 to change member data
Please type 9 to find member data
Please type 10 to exit the system
Enter your selection Here: 9

***Find member data***
Please enter the search keyword: Ruby

The searching keyword is "Ruby"...

We cannot find the keyword.

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
```

Lab 14.4
Reversing String

- In this lab, you are asked to write a **object-oriented program** for reversing a given string
- This program asks the user to enter a string, and reverses the string.
- Finally, showing the reversed string in the program.
- The string includes spaces, numbers, alphabets and special symbols.



Lab 14.4
Reversing String

- There is a limit on the type of variable and the functions in the program:

```
class reverse_str
{
public:
   void set_string(string);
   void print_string();
   void reverse_string();

private:
   string str0;
};
```

- std::reverse is prohibited!!!!



Lab 14.4
Reversing String

- The screen dialog should appear as follows:
- Example 1:

```
Please enter a string:
abcdefgh123456789!@#$%

After reversed, the string is:
%$#@!987654321hgfedcba

Process returned 0 (0x0)   execution time : 13.023 s
Press any key to continue.
```


Lab 14.4
Reversing String

- The screen dialog should appear as follows:
- Example 2:

```
Please enter a string:
1 2 3 4 5 6 7 8 9 / * -

After reversed, the string is:
- * / 9 8 7 6 5 4 3 2 1

Process returned 0 (0x0)   execution time : 11.451 s
Press any key to continue.
```



Evaluation



- Lab14.1(30%)
  - 5 testcase, 6% each.
- Lab14.2(30%)
  - 5 testcase, 6% each.
- Lab14.3(30%)
  - 5 testcase, 6% each.
- Lab14.4(20%)
  - 5 testcase, 4% each.

